SSM - Mental Health 5 (2024) 100302

Contents lists available at ScienceDirect

SSM - Mental Health

journal homepage: www.journals.elsevier.com/ssm-mental-health

Review

The mental health consequences of interpersonal gun violence: A systematic review

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ARTICLE INFO

Handling Editor: Dr A Tsai

Keywords:
Gun violence
Firearm violence
Mental health
Depression
Anxiety
Post-traumatic stress

ABSTRACT

Background: Gun violence is a global public health concern that has significant implications for mental health. This systematic review aims to synthesize the existing literature on the mental health consequences of interpersonal gun violence.

Methods: A comprehensive search was conducted across PubMed, PsycInfo, and Web of Science databases to identify relevant studies published up to December 15, 2023. Studies were included if they documented the mental health impact of interpersonal gun violence. The data extraction process included information on study design, population characteristics, exposure to gun violence, mental health indicators, and measurement instruments.

Results: A total of 58 studies were included in the review. The studies varied in design, population, and mental health indicators assessed. We found that exposure to interpersonal gun violence was significantly associated with increased reports of post-traumatic stress disorder, depressive symptoms, and substance use. We also found that interpersonal gun violence exposure was associated with increased disruptive behaviors and the likelihood of gun carrying especially among adolescents. Further, those closer to the interpersonal gun violence consistently reported more adverse mental health consequences than persons less centrally involved. The review also highlighted limitations in the current literature, including a narrow focus on certain mental health indicators and a paucity of studies from countries other than the US.

Conclusion: The consequences of interpersonal gun violence extend beyond immediate physical injury to include enduring mental health impacts. This review also highlights the need for further research to better understand the impact of gun violence on mental health.

1. Introduction

1.1. Background

Gun violence is a global public health concern. Best available estimates suggest that in 2016 there were approximately 251,000 firearm-related fatalities outside of war zones world-wide, an increase from an estimated 209,000 in 1990 (The Global Burden of Disease 2016 Injury Collaborators, 2018 Injury Collaborators, 2018). Further, there is substantial regional clustering of non-combat firearm mortality, with six countries in the Americas— Mexico, Brazil, Venezuela, Colombia, Guatemala, and the United States (US)— contributing to over half of all global firearm fatalities. Two countries, Brazil and the US, account for 32% of all global firearm-related deaths (The Global Burden of Disease 2016 Injury Collaborators, 2018 Injury Collaborators, 2018).

In 2021, there were over 48,830 gun-related fatalities and 175,459 non-fatal injuries in the US (National Center for Injury Prevention and Control, 2023). While gun violence is not unique to the US, the rate of gun violence in the country is significantly higher compared to other high-income countries. A 2016 study reported that the gun homicide rate in the US was 25.2 times higher than that of other high-income countries (Grinshteyn and Hemenway, 2016). The economic implications of firearm violence in the US are considerable, with an annual cost of around $557 billion—2.6% of its national Gross Domestic Product (Everytown Research & Policy, 2022).

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https://doi.org/10.1016/j.ssmmh.2024.100302
Received 5 August 2023; Received in revised form 17 January 2024; Accepted 14 February 2024
Available online 21 February 2024
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1.2. Instrumentality of firearms in violence

Gun violence includes a range of lethal or non-lethal, accidental or deliberate acts, directed against oneself or others using firearms. The instrumentality of firearms significantly amplifies the severity and consequences of violent acts, distinguishing them from other forms of violence in both lethality and psychological impact (Braga et al., 2020). Firearms have an inherent immediacy and can inflict fatal injuries from a distance, often before a potential victim can react or seek help. This attribute significantly increases the potential for severe physical harm, making firearm-related incidents more likely to result in fatalities compared to other forms of violence (Braga et al., 2020; Weaver et al., 2004). The risk of death in firearm-involved violent encounters is about five times greater than that of other weapons like knives (Weaver et al., 2004). Access to more lethal means (including firearms) is also a significant risk factor for suicide (Miranda-Mendizabal et al., 2019). Further, healthcare costs and resource utilization for firearm-related injuries significantly exceed those for non-firearm violence, reflecting the increased severity and complexity of these injuries (Wolf et al., 2019).

Similar to its higher propensity for physical harm and healthcare costs, the psychological distress from firearm violence is likely greater compared to other forms of violence. There is evidence that victimization involving potentially lethal weapons, such as guns or knives, is associated with increased report of symptoms of mental health problems (Mitchell et al., 2015). Notably, this increased symptom report is likely linked to firearm violence, and not merely a result of poly-victimization. Persons injured by guns are at a significantly higher risk of post-traumatic stress disorder (PTSD), anxiety, depression, and substance use, compared to those injured through other violent means (Kagawa et al., 2018).

Evidence from specific subsets of gun violence, such as mass shootings and police-involved shootings, reveal a substantial mental health burden associated with these incidents (Bor et al., 2018; Lowe and Galea, 2017). However, these types of shootings, while devastating and highly publicized, represent only a small fraction of overall gun violence

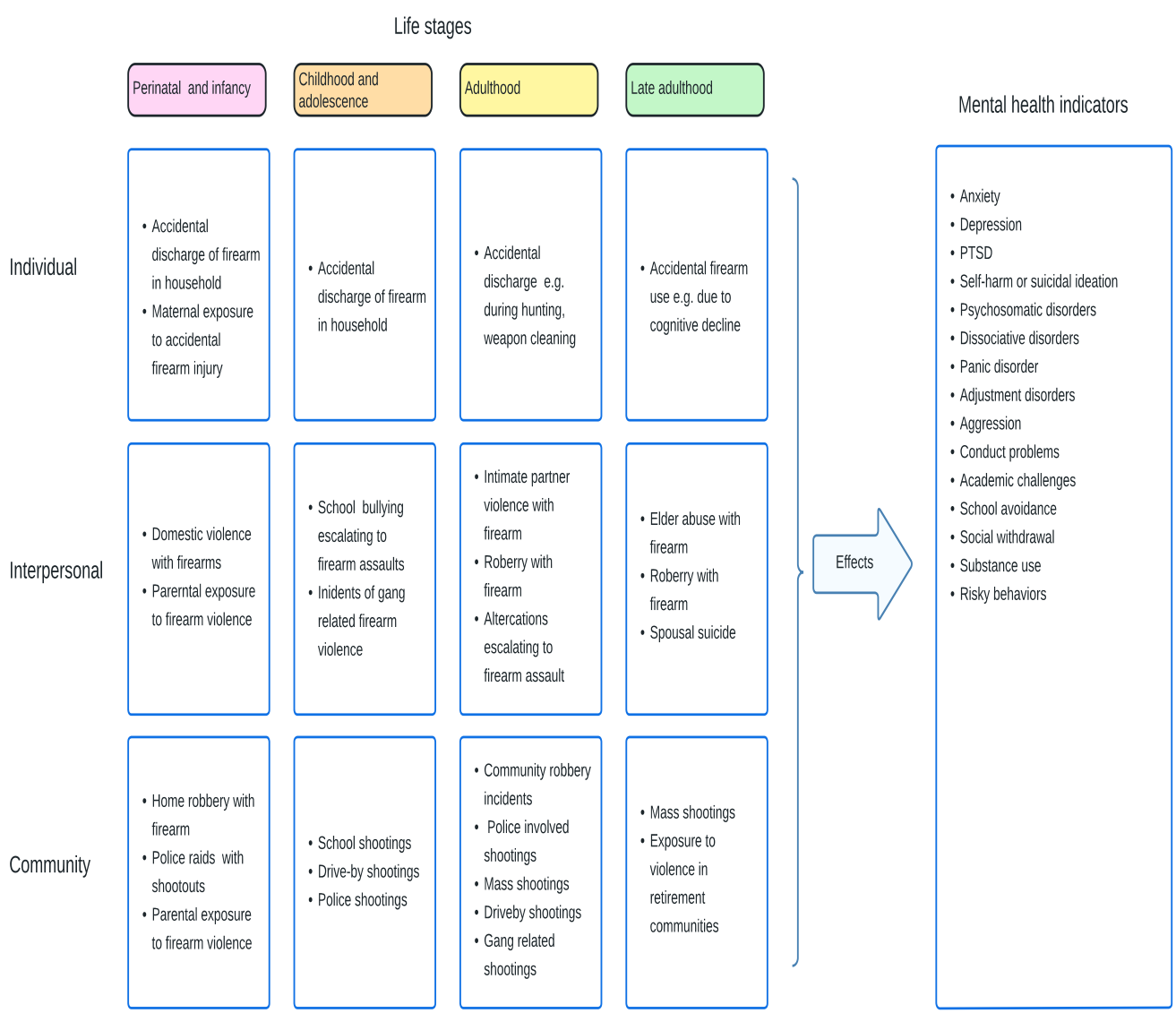

Fig. 1. Integrated life course and socio-ecological model of gun violence and mental health.

cases. In 2021, of the 48,830 gun-related deaths in the US, only 706 (1.5%) were attributed to mass shootings, and an estimated 537 (1%) to police shootings (Gun Violence Archive, 2023; National Center for Injury Prevention and Control, 2023; Pew Research Center, 2023). Exposure to gun violence in the neighborhood may lead to preterm birth, which has effects throughout the life course (Goin et al., 2020; Goin DE et al., 2019). Many more people are affected by individual incidents of firearm violence, such as homicides, assaults, and accidents—either directly as victims, or indirectly within their respective communities (Schumacher et al., 2023). There is therefore the need to closely examine and address other more pervasive forms of gun violence.

1.3. Exposure to gun violence across the life course and diverse socio-ecological contexts

Gun violence victimization occurs at various stages of the life course and within diverse socio-ecological contexts. In exploring the multifaceted nature of gun violence victimization, we adopt a life course approach similar to Fink and Galea (2015) (Fink and Galea, 2015). This approach emphasizes the complex interplay of life experiences, from social to biological factors. Across the life course, the nature of gun violence victimization varies. In the perinatal and infancy stages, this typically involves exposure through accidental firearm discharges at home or during incidents of domestic violence. During childhood and adolescence, victimization often occurs within and around schools, neighborhoods, and homes. Adults are exposed to gun violence in personal conflicts, workplaces, or as a result of criminal activities. In late adulthood, victimization can stem from targeted incidents or broader community violence. We show this conceptual framework as Fig. 1.

At the socio-ecological level, victimization can happen at the individual level, as self-directed acts like suicides and accidental discharges; at the relational/interpersonal level, as intimate partner violence or aggravated assault; or at the community level, through witnessing or hearing about firearm related violence within one's community. However, there is some fluidity with regards to these exposures and how they are often assessed and as such community and interpersonal exposures are often collectively referred to as interpersonal gun violence.

1.4. Mental health consequences

The mental health consequences of gun violence also manifest differently across various life stages. At the perinatal and infancy stages, exposure to accidental firearm discharges or domestic violence incidents can lead to early trauma and developmental disruptions (Shonkoff et al., 2012). During childhood and adolescence—a period marked by increased risk for violence—victimization in schools, neighborhoods, and homes can result in acute stress, anxiety disorders, and impact long-term emotional well-being (Finkelhor et al., 2015). Adults exposed to gun violence may experience heightened risks of PTSD, depression, and substance abuse. Yet, despite the pervasiveness of gun violence throughout the life course, our understanding of its mental health effects remains limited.

Documenting the consequences of gun violence is often subsumed under research on the consequences of violence such as intimate partner violence or community violence. However, different types of violence, including gun violence, have unique impacts on mental health, a perspective often overlooked when gun violence is considered a subset of general violence research (Kennedy and Ceballo, 2014). The historical context in the US has also contributed to the paucity of studies on the mental health impacts of gun violence. Of note is the Dickey Amendment, a provision meant to ensure that federal funding would not be used to advocate for gun control. This amendment and its associated federal funding restrictions have historically curtailed in-depth research into the specific causes and consequences of gun violence (Rostron, 2018). However, recent policy changes and a growing consensus on the necessity for evidence-based approaches to gun violence, have prompted a gradual increase in focused research efforts (Galea et al., 2018).

1.5. Current review

In this systematic review we focus specifically on the mental health consequences of interpersonal gun violence, excluding mass shootings, terrorism, war-time shootings, and police-involved shootings. While equally important to the discourse about guns and their health impacts, our decision to exclude these forms of gun violence is informed by their distinct nature. Mass shootings, while highly publicized and devastating, constitute a small fraction of overall gun violence. Moreover, the mental health consequences of mass gun violence have been recently reviewed (Lowe and Galea, 2017). Similarly, police shootings involve unique dynamics of state authority and public trust. This form of violence, distinct due to its state-sanctioned nature and historical role in enforcing social order, especially against marginalized and minoritized communities, requires a separate focus in understanding its mental health impacts (DeVylder et al., 2020; Haile et al., 2023). Additionally, acts of terrorism and war-related violence are excluded due to their unique political and ideological contexts, which differ significantly from the community and interpersonal context of our review (Charlson et al., 2019; DiMaggio and Galea, 2006). Lastly, firearm suicides—which represent a complex interplay of mental health issues and access to lethal means, including gendered dynamics—are also excluded (Miranda-Mendizabal et al., 2019). This type of gun violence and its impact on the individual and community will require a separate analysis.

The aim of our systematic review, therefore, is to synthesize the expanding body of literature on the mental health consequences of gun violence, particularly focusing on interpersonal violence across the different stages of the life course.

2. Methods

2.1. Search strategy

We searched PubMed, PsycInfo, and Web of Science databases (including sub-databases such as Science Citation Index Expanded Social Sciences Citation Index, and the Arts & Humanities Citation Index). Our search was from inception up to 15 December 2023 and followed Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Moher et al., 2009). The review was registered with PROSPERO (CRD42023417292).

We searched PubMed, PsycInfo, and Web of Science using a combination of PubMed MeSH terms, and keywords related to gun violence, firearms, violence, mental disorders, and mental health. This search strategy was developed in collaboration with a librarian and aimed to capture a wide range of articles that investigated the relationship between gun violence exposure and various mental health indicators. A detailed search strategy is provided in Appendix 1.

This initial search produced a total of 9922 articles (PubMed = 3,438, PsycInfo = 4,293, Web of Science = 2191). We downloaded all the articles and uploaded them to Rayyan (Ouzzani et al., 2016), an online collaborative systematic review platform. We identified 1788 duplicates using the automated duplication detection function in Rayyan systematic review software and hand search. This left us with a total of 8134 records for further screening.

2.2. Inclusion/exclusion criteria

We included articles if they: i) were peer-reviewed; ii) used a quantitative study design (cross-sectional, case-control, quasi-experimental or longitudinal studies); iii) reported on individuals who have been exposed to interpersonal gun violence, either directly or indirectly; and iv) assessed mental health outcomes such as anxiety, depression, PTSD, other trauma-related mental health symptoms. We included

articles published globally.

We excluded case reports, case series, editorials, commentaries, conference abstracts, book chapters, preprint articles, and systematic literature reviews. We further excluded studies on violence that did not report on exposure to gun violence or when such exposures could not be disaggregated from other categories of violence (e.g., studies that reported homicide with a weapon but did not specify types of weapons used). Additionally, we excluded studies that did not report either direct or indirect exposure to actual gun violence incidents. These include studies that primarily focused on concerns or perceptions of gun violence, the presence of firearms in household, or exposure through mass media, social media, or video games. Publications where gun violence was considered as the outcome of mental health disorders (e.g., mental health of perpetrators of gun violence) were also excluded. Last,

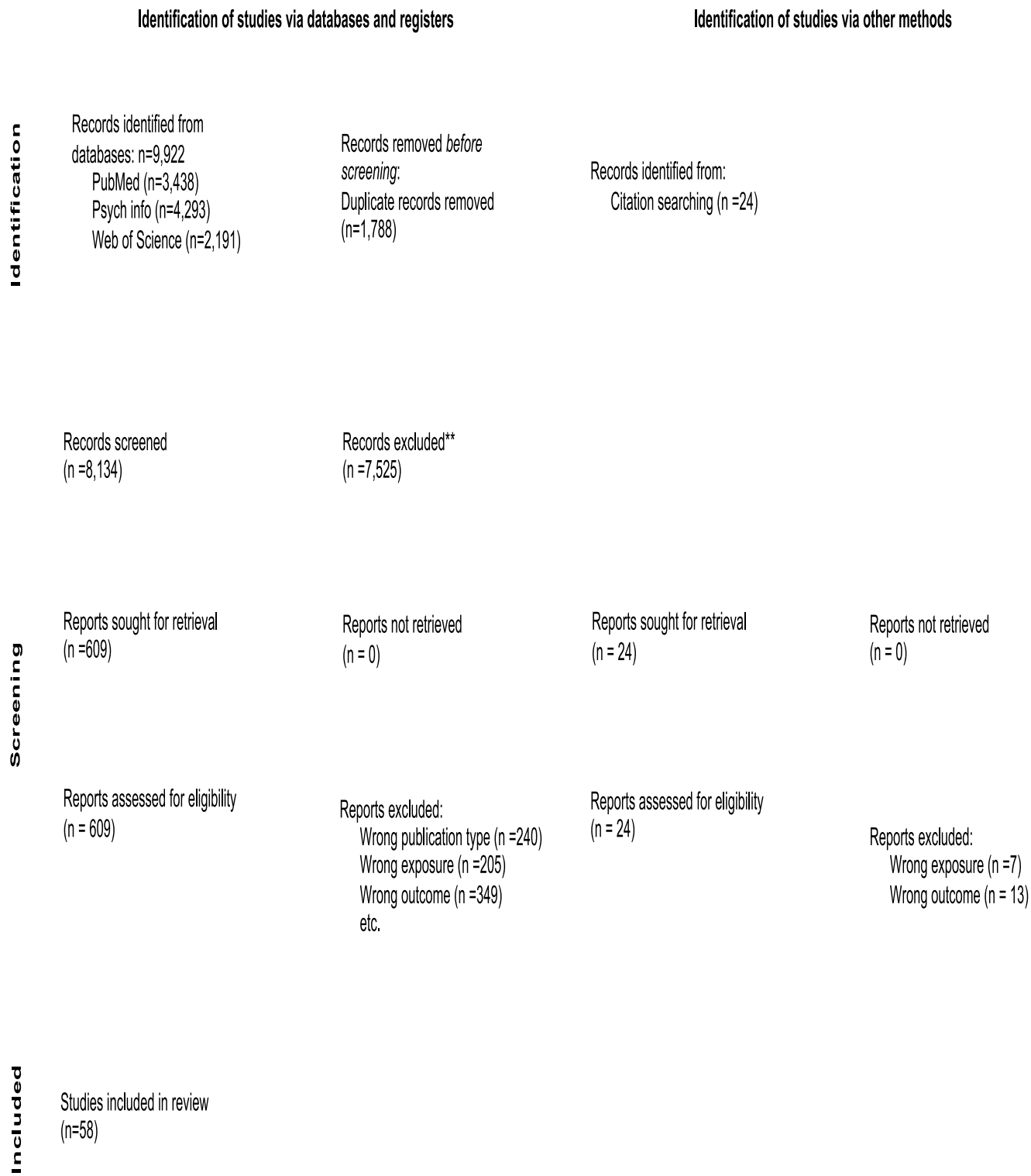

Fig. 2. Study identification, screening, and eligibility test, following the Preferred Reporting Items of Systematic Reviews (PRISMA).

studies on mass shootings, police killings, terrorism and armed conflicts were also excluded.

Two authors (MMA and SFK) independently screened and reviewed records based on title and abstract. This stage was performed using Rayyan systematic review software and conflicts were resolved through consensus. 7525 studies were excluded at this stage of the review, leaving 609 potentially eligible articles. We further excluded records because they either did not report mental health indicators (n = 349); did not report outcomes based on exposure to gun violence (n = 205); or for other reasons e.g. not quantitative studies (n = 240). Some articles had more than one reason for exclusion. For brevity, we list only the main causes for exclusion here. In addition to the database search, we also manually examined the reference lists of the articles identified through our initial search. This hand-searching process was conducted to ensure we did not overlook any relevant studies that might not have been captured in the database search. Through this process, we identified four additional articles that met our inclusion criteria. This left us with 58 articles for data analysis and quality assessment. In line with the principles outlined in the PRISMA guidelines (Moher et al., 2009), we present the details of the study selection flow in Fig. 2.

2.3. Data extraction, analysis, and quality assessment

We extracted the following study parameters into a summary Excel table: Author(s) and year of publication, country/location, study period, study design, sampling method, data collection method, population, sample size, measurement instrument for exposure, type of gun violence exposure, comparator/control, mental and behavioral health indicator (s) assessed. Our decision to report our outcomes as "indicators" is based on the population health approach which recognizes that the health states exist on a continuum (Keyes and Galea, 2016). This approach is particularly relevant in the study of mental health across the life course where mental health conditions among children and adolescents often start to manifest as externalizing or internalizing behaviors. Additionally, we extracted the measure or instrument used to assess outcome, key findings including proportions, means and standard deviations (SD), and effect sizes such as odds ratios (OR) with 95% confidence interval (CI) where available.

We appraised the methodological quality of cohort and cross-sectional studies using the National Heart, Lung, and Blood Institute (NHLBI) Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies (National Heart Lung and Blood Institute & National Institutes of Health, 2014). We used another tool, the NHLBI, Quality Assessment Tool for Before-After studies to assess pre-post studies with no control group (National Institutes of Health, 2014). Two reviewers (MMA and SFK) independently assessed the quality of the studies. Disagreements were resolved through discussion and in cases where these disagreements were unresolvable, a third independent reviewer (SMA) was asked to clear out any ambiguity. Each study was assessed based on a set of criteria, with responses categorized as "yes", "no", "not applicable", or "cannot be determined". Based on the NHLBI quality grading criteria, studies with a score of 10–14 were considered of "Good" quality, those with a score of 6–9 were considered "Fair", and those with a score of 0–5 were considered "Poor" quality.

We did not do a meta-analysis because there was significant methodological and clinical heterogeneity among study populations, designs, outcome measure instruments, and results in the examined studies. Instead, we performed a Synthesis Without Meta-analysis (SWiM) (Campbell et al., 2020). SWiM is an effective strategy for aggregating studies with substantial methodological and clinical heterogeneity as it accommodates diverse study designs, populations, and outcomes.

2.4. Funding source

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

3. Results

We present our findings systematically, starting with an overview of the study characteristics, followed by a quality assessment of the included studies and an overview of the mental health indicators. We then present our findings, beginning with mental health indicators for children and adolescents before proceeding to those for adults.

3.1. Overview of study characteristics

Fifty-five (n = 55) out the 58 studies in our review were conducted in the US. The three studies from outside the US were from Turkey, Brazil, and Canada. The included studies were largely longitudinal (n = 30), cross-sectional studies (n = 27), along with one case-control study. All the studies were published between the years 1987 (Pynoos, 1987) to 2023 (Galovski et al., 2023). Sample sizes in our included studies ranged from 24 (Ganpo-Nkwenkwa et al., 2023) to 114,526 (South et al., 2021). The total sample size across all the included studies was 414,891.

3.2. Quality assessment of included studies

Overall, the quality score for the sample of the 58 included studies ranged from 6 to 12, with a higher score indicating better methodological quality. Out of the 58 studies, 21 were rated as "Good" (scoring between 10 and 12), 37 were rated as "Fair" (scoring between 6 and 9), and none was rated as "Poor". A common methodological limitation across many of the included studies was the lack of justification for the sample size as most studies did not provide a clear rationale for their chosen sample size. Moreover, only a few studies explicitly reported blinding of outcome assessors to the exposure status of participants. For most studies, this was not mentioned. Fewer than half of the studies (28 out of 58) examined different levels or amounts of exposure to gun violence in relation to mental health indicators. While we did not exclude studies based on the quality assessment, it is important to note that our synthesis focused more on studies of higher quality for mental health incomes where such studies are available. Details of quality assessment and scores are in Appendix 2.

3.3. Mental health indicators

We present the mental health indicators across life stages life stage—childhood and adolescence (n = 35) and adulthood (n = 32). These mental health indicators are categorized under mood disorders (n = 18), anxiety disorders (n = 11), substance use (n = 9), trauma and stress related disorders (n = 26), disruptive, impulse control, and conduct disorders (n = 19), and other types of unclassified mental health disorders (n = 14). In presenting our findings, we first provide where available, the burden (e.g., prevalence/incidence/mean scores) followed by the measures of effect (e.g., OR, RR) reported. Table 1 highlights the key findings of all included studies. Figs. 3 and 4 show the Tree map of these mental health indicators and their distribution by children and adolescents and adults respectively.

3.3.1. Children and adolescents (n = 35)

Thirty-five studies reported a range of mental health indicators among children and adolescents. The most common mental health indicators studied were disruptive, impulse control, and conduct disorders (n = 14), followed by anxiety (n = 9), mood disorders (n = 9), and trauma and stress related disorders (n = 8). Only three studies reported on substance use among this group. See Fig. 4.

3.3.2. Mood disorders (n = 9)

Nine of the included studies reported on mood disorders among children and or adolescents (Borg et al., 2021; Buggs et al., 2022; Leibbrand et al., 2020; Lennon et al., 2023; Magee et al., 2022; Magee et al., 2022b; Shulman et al., 2021; Slovak and Singer, 2001) Of these

Table 1
Study characteristics and key findings for 58 included studies.

Author(s) and year of publication	Study design [a]	Population [c]	Sample size [d]	Mental health indicators [e]	Main findings	Quality score
Allwood et al. (2023) (Allwood et al., 2023)	Cross-sectional study	College students aged 25 years and younger (Adults)	894	PTSD	Gun victimization was associated with a higher likelihood of PTSD. Knife victimization had no significant effect. Significant interaction between gun victimization and race in relation to PTSD symptoms. Gun victimization: OR = 2.30; Interaction between gun victimization and race: OR = 0.55, p = 00.025	8
Aubel et al. (2021) (Aubel et al., 2021)	Cross-sectional study	Adults residing in California	2558	Distress levels, problems with job/school, problems with family/friends, and gun purchasing intentions	Violence involving a firearm was associated with a higher prevalence of severe distress compared to violence involving other forms of weapon (69.1% vs 11.8%). However, social functioning problems were less among those exposed to firearm violence compared to those exposed to violence with other forms of weapon (67.2% vs 30.7%).	7
Bailey et al. (2013) (Bailey A et al., 2013)	Cross-sectional study	Adult women	48	Traumatic stress symptoms	Social support and cognitive appraisal appeared to mediate the relationship between the level of stress experienced by mothers bereaving a child to gun violence and their resilience.	6
Beardslee et al. (2018) (Beardslee et al., 2018)	Longitudinal study [b]	Male juvenile offenders (14–19 years)	1170	Gun carrying behavior	Gun violence exposure increased the odds of gun carrying by 43%.	11
Beardslee et al. (2021) (Beardslee et al., 2021)	Longitudinal study	Male youth enrolled in the Crossroads Study	1216	Gun carrying behavior	Youth were more likely to carry guns in years they experienced gun violence (odds ratio [OR] = 4.25) than in years exposed to non-gun violence (OR = 2.15). The likelihood of carrying a gun was 4.3 times greater when witnessing or experiencing gun violence, but not significantly higher for exposure to other serious violence without guns.	12
Bingenheimer et al., 2005	Longitudinal study	Adolescents residing in Chicago	1517	Serious violence perpetration	Those with exposure to firearm violence had a higher likelihood (adjusted OR = 2.47) of committing serious violence compared to those not exposed, within the same group.	11
Borg et al. (2021) (Borg et al., 2021)	Cross sectional study	Urban (Detroit) and non-urban residents (aged 6–17 years)	64	Anxiety and depression	Urban youth who reported hearing gunshots had 13 times higher odds for clinical anxiety compared to those without such an experience.	8
Buggs et al. (2022) (Buggs et al., 2022)	Longitudinal study	Youths (aged 2–17 years)	3086	Anxiety and depression	Spatial proximity to firearm homicide significantly increases depression symptoms in youth, especially in boys.	12
Büken et al. (2009) (Büken et al., 2009)	Cross-sectional study	Adolescents exposed to firearm violence	791	Stress, aggression, alcohol use and smoking	30.9% of students exposed to some of form of gun violence (e.g., shot with a firearm, threatened with a gun or heard shooting each day) reported smoking and alcohol use.	8
Comer & Connolly (2023) (Comer BP & Connolly EJ, 2023)	Longitudinal study	U.S. youth born between 1980 and 1984 (middle adolescence to adulthood)	6748	Gun carrying behavior	Early life exposure to gun violence is associated with an increased risk for illegal handgun carrying during adolescence.	10
Ehrlich et al. (2022) (Ehrlich et al., 2022)	Matched case-control study	Children aged 3–17 years with firearm or motor vehicle crash injuries	5141	New mental health diagnoses post-injury	Among children who suffered a firearm injury, about 35% receive a new mental health diagnosis within a year following the injury, compared to 26% of those who sustained a motor vehicle crash (MVC) injury. Children with firearm injuries had 1.5 times the odds of having a new mental health diagnosis compared to those with MVC injuries.	10
Forster et al. (2015) (Forster et al., 2015)	Cross-sectional study	7th and 8th-grade students	164	Aggression	Witnessing gun violence increased aggression (Incident Risk Ratio = 1.36, 95% CI:1.20–1.54).	10
Fullerton et al. (2015) (Fullerton et al., 2015)	Cross-sectional study	Adults living in Washington DC during the weeks of the 2002 sniper attacks	1238	PTSD, depression, and alcohol use.	About 8% of the participants met criteria for probable PTSD, 22% reported mild to severe depression, and 4% reported increased alcohol use during the aftermaths of the attacks.	9

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Table 1 (*continued*)

Author(s) and year of publication	Study design [a]	Population [c]	Sample size [d]	Mental health indicators [e]	Main findings	Quality score
Galovski et al. (2023) (Galovski et al., 2023)	Cross-sectional study	Veterans (aged 18–50 years)	3049	PTSD and depression.	Neither depression nor PTSD symptoms were linked to exposure to gun violence in this study.	7
Ganpo-Nkwenkwa et al. (2023) (Ganpo-Nkwenkwa et al., 2023)	Longitudinal study	Adults aged less than 18 years at the time of firearm injury.	24	Global mental health, alcohol use, severity of substance use and PTSD	A high prevalence of probable PTSD reported among patients with firearm injury (41.7%).	7
Gard et al. (2022) (Gard et al., 2022)	Longitudinal study	Youth living in US cities.	3444	Youth delinquency	For every incidence of fatal gun violence within 500 m of a child's home, there was a 9.6% increase in youth delinquency.	10
Greenspan & Kellermann (2002) (Greenspan and Kellermann, 2002)	Longitudinal study	Adult gunshot injury victims (aged 18–64 years) in metropolitan Atlanta area.	60	PTSD, decline in physical and mental health status	Majority of patients experienced a decline in physical and mental health status with over 80% reporting moderate or severe symptoms of post-traumatic stress.	10
Grieger et al. (2003) (Grieger et al., 2003)	Cross-sectional study	Employees of a large tertiary care military hospital	382	Acute stress disorder, depression, and increased alcohol use	Prevalence of acute stress disorder was 6%, increased alcohol use was 3%, and depression was 8%.	8
Haasz et al. (2023) (Haasz et al. (2023))	Longitudinal study	Children and adults injured in firearm incidents or MVCs	48	Long-term physical and mental health outcomes	No significant difference in depression scores between the firearm injury patients and MVC injury patients.	7
Herrera-Escobar et al. (2021) (Herrera-Escobar et al., 2021)	Longitudinal study	Adults (aged 18–64 years)	317	PTSD and mental health-related quality of life	Firearm injury survivors had poorer mental health-related quality of life and greater risk for PTSD compared to motor vehicle crash survivors.	10
Hoffmann et al. (2023) (Hoffmann et al., 2023)	Longitudinal study	Children (aged 5–17 years)	2613	Mental health (MH) service utilization	36.7% used MH services within 6 months of injury, 39.5% of these were first-time users.	8
Hsu et al. (2020) (Hsu et al., 2020)	Cross-sectional study	Young adults experiencing homelessness aged 18–24 years	1426	Suicidal ideation	No relationship between either direct or indirect victimization and suicidal ideation.	8
Joseph et al. (2019) (Joseph et al., 2019)	Cross sectional study	Adult patients with firearm injury-related admissions	100,702	PTSD	The overall incidence of readmission with acute stress disorder/PTSD was 9%. Factors such as weapon type and substance abuse influenced the odds of readmission.	11
Kagawa et al. (2020) (Kagawa et al., 2020)	Cross-sectional study	Persons aged 12 years and older.	5698	Severe distress and daily functioning problems	Distress and functioning problems were higher in victims of firearm violence compared to victims of non-firearm victimization.	6
Lei et al. (2023) (Lei et al., 2023)	Longitudinal study	Children and adolescents with firearm injury and their family members	532	Mental health care cost	Increased mental health care for the firearm injured youth within the first three months post-injury.	10
Leibbrand et al. (2020) (Leibbrand et al., 2020)	Longitudinal study	Adolescents (aged 15 years at time of study).	2823	Externalizing behavior problems, anxiety, depression	Proximity to gun homicide incidents was associated with higher indices of behavior problems, anxiety, and depression.	10
Leibbrand et al. (2021) (Leibbrand et al., 2021)	Longitudinal study	Mothers in 20 US urban cities	4587	Depression	Mothers who witnessed a shooting had a 32–60% higher risk of depression compared to mothers who had not.	11
Lennon et al. (2023) (Lennon et al., 2023)	Cross-sectional study	Children aged 2–17 years with firearm injury in Chicago.	989	Anxiety, sadness, isolation, difficulty concentrating, and aggression.	Mental health symptoms were higher among children exposed indirectly (31%) and directly (68%) to firearm violence compared to those unexposed (7%).	9
Lumbard et al. (2023) (Lumbard DC et al., 2023)	Longitudinal study	Patients aged ≥14 years with assault-related firearm injury	20,666	Unplanned readmission rates, causes for readmission	4% of unplanned readmissions were due to mental health issues, with PTSD contributing to about 10% of these within 6 months post-firearm injury.	9
Lyons et al. (2019) (Lyons et al., 2019)	Longitudinal study	Adults with assault or unintentional firearm injury.	163	PTSD, depression, social support, happiness, physical and mental health, Alcohol Use Disorders	Mean PTSD score-34.0 (17), PHQ-8 depression score- 6.2 (6), SF12 Mental Score- 48.8 (13), Alcohol use score-3.0 (3); Most patients reported no change in firearm exposure (80.4%);	9
M. E. Smith et al. (2020) (M. E. Smith et al., 2020)	Cross-sectional study	Adult residents of four U. S. cities	1615	Psychological distress, depression, and psychosis-like experiences	Persons exposed to a fatal gun violence were more likely to experience severe psychological distress and depression. They also had greater risks for suicidal ideation and psychosis-like experiences.	7
Magee et al. (2022) (Magee et al., 2022)	Longitudinal study	Nonfatal shooting victims	2681	Depression, anxiety, disruptive behavior disorder and substance use	Mental health and substance use disorder rates significantly increased after nonfatal firearm injury. Prevalence rates for depression, anxiety, and substance use disorders significantly increased among adult survivors after the incident. Disruptive behavior disorder increased among youth survivors.	9

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Table 1 (*continued*)

Author(s) and year of publication	Study design [a]	Population [c]	Sample size [d]	Mental health indicators [e]	Main findings	Quality score
Magee et al. (2022a) (Magee et al., 2022)	Longitudinal study	Nonfatal shooting victims and their family members	2838	Stress, anxiety, depression, mood disorders, disruptive behavior disorders and substance use	The overall number of mental health claims increased for family members after the shooting. About 6% of survivors and family members were diagnosed with a new mental health disorder after the shooting. Mental health diagnoses slightly decreased among youth survivors.	9
Mello-silva et al. (2012) (Mello-Silva et al., 2012)	Cross-sectional study	Adult victims of interpersonal firearm violence.	717	Post-traumatic stress symptoms	About 60% of victims of interpersonal firearm violence had PTSD symptoms.	7
Nehra et al. (2021) (Nehra et al., 2021)	Longitudinal study	Adult survivors of firearm injury	635	PTSD and firearm carriage	Firearm injury admission was linked to increased post-injury firearm possession and significantly elevated PTSD symptoms during the year following a firearm injury admission.	12
Oddo et al. (2021) (Oddo et al., 2021)	Longitudinal study	Youth survivors of firearm injury (aged <18 years)	2178	New mental health diagnosis one year after a firearm injury	Nonfatal firearm injuries were associated with a new mental health diagnosis post-injury among victims. 25% of victims had a new mental diagnosis in the year following a firearm injury.	12
Phillips et al. (2020) (Phillips et al., 2020)	Longitudinal study	Pediatric patients with a history of gunshot injury	308	30 days and 1 year follow-up for psychiatry appointments	At 30 days and 1-year post-injury, 14.0% of patients had a psychiatric follow up. Rates of follow up visits for teenagers and children were similar.	7
Pulcini et al. (2021) (Pulcini et al., 2021)	Longitudinal study	Children aged 0–18 years with firearm injuries	3296	Mental healthcare encounters and expenditures	Overall mental health expenditure dropped by $346,029, with significant changes across subgroups. The study found varied mental healthcare utilization changes post-injury, with increases in low prior expenditure youth and decreases in high prior expenditure youth.	7
Pynoos (1987) (Pynoos, 1987)	Cross-sectional study	Children exposed to ta fatal sniper attack	159	Traumatic stress symptoms	About 38.4% of children exposed to the incident had moderate or severe PTSD symptoms. 22% reported mild PTSD symptoms. The severity and type of PTSD symptoms were closely related to the level of exposure to the shooting. There was statistically significant increase in PTSD scores with increased level of exposure.	8
R. N. Smith et al. (2018) (R. N. Smith et al., 2018)	Longitudinal study	Black male survivors of firearm injury	139	PTSD and depression	Patients with retained bullets had more severe depressive symptoms compared those without it. This was significant even after adjusting for sociodemographic and injury related factors (severity and number of injury) ($\beta = 3.52$; $p = 0.017$). No significant difference in PTSD scores between the two patient groups.	10
Reese et al. (2012) (Reese et al., 2012)	Cross-sectional study	Adolescents and adult survivors of urban trauma.	307	Traumatic stress symptoms	Gunshot wound victims were 13 times more likely than fall victims and twice as likely as MVC victims to test positive for PTSD.	9
Reid et al. (2017) (Reid et al., 2017)	Longitudinal study	Male youths aged 14–19 years who had been found guilty of a serious criminal offense	1170	Gun carrying behavior	Witnessing violence was a significant predictor of gun carrying at all 4 stages of assessment, with increased odds ranging from 1.43 to 1.59 for each unit increase in exposure.	12
Schulden et al. (2006) (Schulden et al., 2006)	Cross-sectional study	Adults and elderly exposed to sniper attack.	1205	Traumatic stress symptoms	Women living within 5 miles of an incident were over four times more likely to report higher traumatic stress symptoms compared to those living farther away (OR = 4.2). No similar association was found for men.	8
Self-Brown et al. (2011) (Self-Brown et al., 2011)	Cross-sectional study	Parents of children in Washington DC or two adjacent counties during the sniper shootings	1205	Parent-reported child psychological distress	Various factors, such as child age, gender, prior trauma, routine disruption, and parent traumatic stress symptoms, were associated with a higher number of distress symptoms in children exposed to sniper shootings.	8
Semenza & Stansfield (2021) (Semenza and Stansfield, 2021)	Longitudinal study	Adults living in 100 large US cities.	14,854	Mental health symptoms	Worsening in overall neighborhood health including mental health among people living in communities affected by gun violence.	9
Sheley & Brewer (1995) (Sheley and Brewer, 1995)	Cross-sectional study	Adolescents of suburban youth.	418	Gun carrying behavior	Girls threatened with a gun had greater likelihood of gun ownership and carrying. For boys, attending events where guns	7

(*continued on next page*)

Table 1 (*continued*)

Author(s) and year of publication	Study design [a]	Population [c]	Sample size [d]	Mental health indicators [e]	Main findings	Quality score
					were fired, and being threatened with a gun increase the likelihood of gun carrying.	
Shulman et al. (2021) (Shulman et al., 2021)	Longitudinal study	Male, justice-involved adolescents	1216	Depression, anxiety, and aggression	Gun violence exposure was significantly associated with increases in reactive aggression, proactive aggression, and anxiety. There was no association with depressive symptoms.	11
Slovak (2002) (Slovak, 2002)	Cross-sectional study	Public school students in grade 3-8	549	Trauma and stress related symptoms	Gun violence exposure was associated with higher levels of anger, dissociation, post-traumatic stress, total trauma, and aggressive behaviors.	8
Slovak & Singer (2001) (Slovak and Singer, 2001)	Cross-sectional study	Students attending a rural students assistance program	171	Trauma and stress related symptoms	After adjusting for demographics factors, gun violence exposure explained 5.9% of variation of overall trauma.	9
Song et al. (2022b) (Song et al., 2022a)	Longitudinal study	Survivors, family members, adults and children	18,087	Pain diagnoses, psychiatric disorders, substance use disorders	Health care costs, usage, and psychiatric disorders significantly increased after nonfatal firearm injuries. There was a 51% increase in psychiatric disorders, 85% increase in substance use disorders and,12% increase in psychiatric disorders among family members.	7
South et al. (2021) (South et al., 2021)	Longitudinal study	Residents living within the selected zip codes of Philadelphia, PA	114,526	Anxiety, depression, and substance use.	No relationship between gun violence and anxiety or substance use found in this study. Greater odds for depression when the analysis was limited to multiple shooting incidents occurring within the previous six months and within a half-mile radius.	9
Teplin et al. (2021) (Teplin et al., 2021)	Longitudinal study	Youth in a juvenile detention center	1829	Not directly specified	Adolescents threatened or injured by firearms had higher odds perpetrating firearm violence later in adulthood.	10
Timmer-Murillo et al. (2023) (Timmer-Murillo et al., 2023)	Cross-sectional study	Adults with firearm injuries	87	PTSD, depression, and anxiety	PTSD severity increased; anxiety and depression levels increased but stress levels remained normal for injured victims.	8
Turner et al. (2019) (Turner et al., 2019)	Cross-sectional study	Youth (aged 2–17 years)	630	PTSD	Both direct and indirect exposure to firearm violence were strongly associated with post-traumatic symptoms in children aged 10–17 years.	7
Vaddiparti et al. (2016) (Vaddiparti K et al., 2016)	Cross-sectional study	Women with substance use.	586	Problem gambling	Problem gambling was associated with gun violence exposure among women with substance use.	8
Vasan et al. (2021) (Vasan et al., 2021)	Cross-sectional study	Children living in 12 zip codes surrounding a large tertiary care pediatric hospital in Philadelphia, PA	54,341	Mental health-related emergency department (ED) utilization	Children living within one-eighth of a mile of a shooting had greater odds of having a mental health-related ED visit up to 60 days after the shooting. Mental health-related ED visits increased when children were exposed to multiple shootings within a 60-day period.	7
Vella et al. (2020) (Vella et al., 2020)	Longitudinal study	Adult patients (aged ≥18 years)	183	Global mental health, alcohol and substance use, and PTSD	Participants showed poorer global mental health scores compared to the US population. While they had better scores in terms of alcohol use and severity of substance use, the combined use of alcohol and substance increased by 13.2% post-gunshot wound (GSW).	8
Weigend Vargas and Hemenway (2021) (Weigend Vargas and Hemenway, 2021)	Cross-sectional study	Adults and children.	18,275	Emotional and physical symptoms	Gun-related crime victims were more likely to report emotional (OR = 1.74, 95% CI = 1.50–2.02) and physical symptoms (OR = 1.37, 95% CI = 1.19–1.60) compared to victims of other weapon related crime.	9

[a] Study design categorizes the research design used in each study.
[b] Both cohort and prospective studies are categorized as longitudinal studies.
[c] Population describes the demographic groups studied (e.g., age, race, region).
[d] Sample size is the number of persons or individual records included in the study.
[e] Mental health indicators' lists the specific mental health conditions or symptoms examined in each study.

nine studies, six reported a significant increase in depressive symptoms with exposure to gun violence.

In a cohort study by Magee et al. (2022) that examined prevalence rates for depression among youth family members of non-fatal shooting survivors, the authors found a significant increase in depression and mood disorder from about 1.78% 12 months preceding the shooting to 2.54% 12 months after the shooting (Magee et al., 2022).

Buggs et al. (2022) found a statistically significant relationship

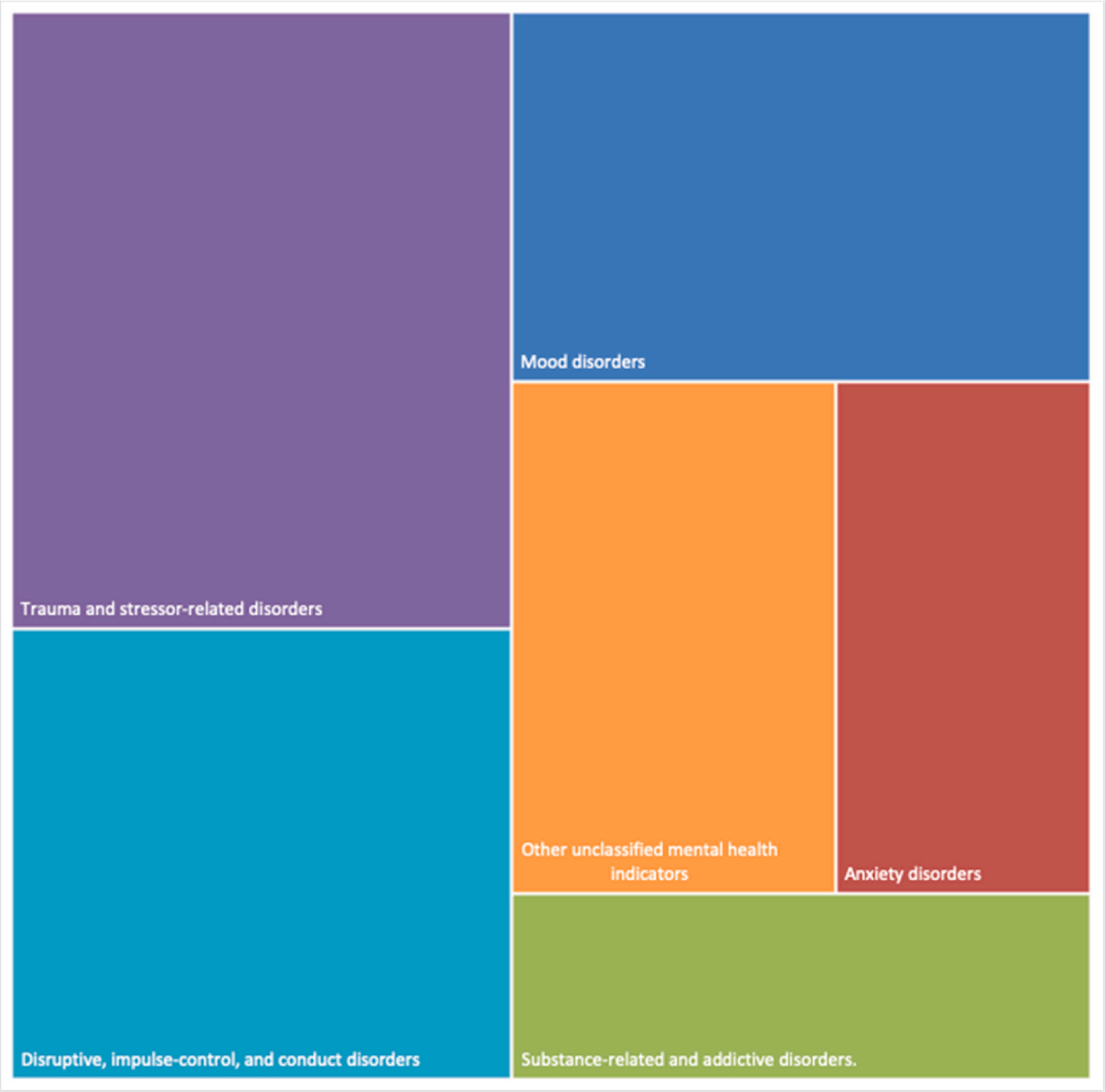

Fig. 3. Tree map of mental health indicators included in review.
Note: Some articles reported multiple mental health indicators, hence the numbers exceeding the total number of included studies.

between spatial proximity to a gun violence incidence within the past year and depression that was stronger among boys (Buggs et al., 2022). Leibbrand et al. (2020) found a similar relationship among girls rather than boys. The authors reported that girls living within a 1-mile radius of a gun homicide had 24% and 16% higher depression scores, compared to those living/schooling beyond a 1-mile radius from a homicide incident (Leibbrand et al., 2020). In a cross-sectional study with a sample of urban and non-urban children (aged 6–17 years) in a US city, Borg et al. (2021) found that urbanicity significantly predicted depression and accounted for some variation in depression symptoms (Borg et al., 2021).

In comparing depression between firearm injury patients and motor vehicle patients, a retrospective cohort study among patients under 18 years of age found no significant difference in depression between the two groups (Haasz et al., 2023). Even though both groups had elevated scores, the difference in mean depression scores between firearm injury patients 58.9 (15.4) and motor vehicle patients 47.0 (7.4) was not significant. Similarly, in a five year-cohort study on male justice-involved adolescents, Shulman et al. (2021) found no relations between exposure to gun violence and depressive symptoms (Shulman et al., 2021).

3.3.3. *Anxiety disorders (n = 9)*

In a longitudinal study among survivors of non-fatal shooting and their families, Magee et al. (2022) reported a significant increase in diagnoses of generalized anxiety disorder among youth family members (pre: 0.9% vs. post: 2.8%) (Magee et al., 2022). Another longitudinal study, using a sample of 1216 male adolescents examined the impact of mixed gun violence exposure— "shot, shot at, and saw someone shot at"—on anxiety and other mental health indicators. After adjusting for potential confounders such as other types of violence, institutionalization, neighborhood disorder, conduct behavior and grief, the authors found that periods of gun violence exposure were significantly associated with more anxiety symptoms ($\beta = 0.12$, 95% CI = 0.01, 0.21) (Shulman et al., 2021). Borg AB et al. (2021) found that urban youth who reported hearing gunshots had 13 times higher odds for clinical anxiety compared to those without such an experience. Magee et al. (2022) found an increase in anxiety diagnosis among youth family members of survivors of non-fatal shooting. However, the authors did not find a statistically significant increase in diagnoses of generalized anxiety disorder among youth survivors (pre: 1.8% vs. post: 3.3%, $p > 0.05$) (Magee et al., 2022).

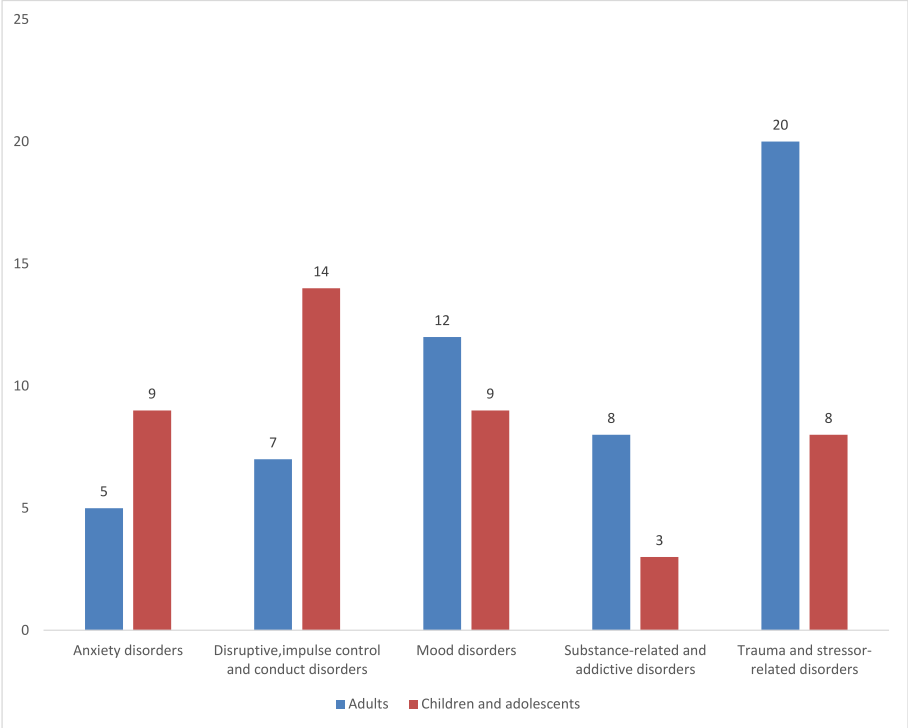

Fig. 4. Distribution of mental health indicators by life stages in included articles (n = 58).
Note: Some articles reported on both children and adolescents and adults, so numbers do not add up to 58.

Similar to their finding on the relationship between community gun homicide and mood disorders, Leibbrand et al. (2020) reported that girls living within a 1-mile radius of a gun homicide had 16% higher anxiety scores, compared to those living/schooling beyond a 1-mile radius from the homicide incident (Leibbrand et al., 2020). This relationship was significant even after adjusting for broader patterns of crime and household and contextual disadvantage. However, Buggs et al. (2022a, b) found no statistically significant relationship between proximity to firearm homicide exposure and anxiety symptoms (Buggs et al., 2022).

3.3.4. *Substance-related and addictive disorders (n = 3)*

Only three of our included studies assessed substance use among children and adolescence (Büken et al., 2009; Magee et al., 2022; Song et al., 2022a). In one of the few studies outside of the US included in our review, Büken et al. (2009) reported that 30.9% of students exposed to some of form of gun violence (e.g., shot with a firearm, threatened with a gun or heard shooting each day) reported smoking and alcohol use (n = 765). Conversely, Magee et al. (2022a) reported a slight decrease in substance use among youth survivors of non-fatal shooting (pre-injury 4.6% vs post-injury 3.6%) (Magee et al., 2022). However, they also reported a modest increase in diagnosis of substance use disorders (Pre: 0.64 vs. Post: 0.76) among youth family members. Although the Song et al. (2022b) study reported a significant increase in substance use disorder up to a year after sustaining a firearm injury, this was not stratified by age (Song et al., 2022a).

3.3.5. *Trauma- and stressor-related disorders (n = 8)*

Eight studies reported on trauma and stressor related disorders among children and adolescents (Allwood et al., 2023; Büken et al., 2009; Kagawa et al., 2020; Pynoos, 1987; Reese et al., 2012; Slovak, 2002; Slovak and Singer, 2001; Turner et al., 2019).

In a study among children and adolescents (2–17 years), Turner et al. (2019) found both direct and indirect exposure to firearm violence to be strongly associated with post-traumatic symptoms in children aged 10–17 years. For younger children (2–9 years), witnessing gun violence and hearing gunshots significantly predicted post-traumatic symptoms, with β coefficients of 0.19 and 0.17 respectively. Among those aged 10–17 years, witnessing gun violence remained a significant predictor of post-traumatic symptoms (β = 0.18) even after considering poly-victimization, while the effect for direct gun violence exposure diminished (Turner et al., 2019). Büken et al. (2009) also reported hearing firearm shootings as the most important stress inducing factor for adolescents (Büken et al., 2009).

Three studies compared risk of post-traumatic stress among victims of firearm injury with that of other forms of injuries (e.g., knife or motor vehicle crash). A hospital-based study examined the associations among physical assaults, weapons-related victimization, and PTSD symptoms. The study authors reported that experiencing gun victimization was associated with more than twice the likelihood of meeting the criteria for clinically significant PTSD (OR = 2.3) compared to victimization without a weapon. Knife victimization was not associated with a significantly greater risk (Allwood et al., 2023). A screening study that piloted that the use of Primary Care PTSD (PC-PTSD) screening tool among patients which included children accessing care at an urban trauma center, reported that firearm injury victims were more likely compared to fall and motor vehicle crash victims to screen positive for PTSD on the instrument (Reese et al., 2012). However, the findings from this study were not stratified by on age. Conversely, in a national sample that compared individuals that were victimized with a firearm to those victimized with another weapon or no weapon, Kagawa and colleagues found no statistical difference in PTSD levels between children adolescents exposed to firearm victimization compared to those victimized without a firearm (Kagawa et al., 2020).

3.3.6. *Disruptive, impulse-control, and conduct disorders: (n = 13)*

These categories of disorders were the most reported among children and adolescents in our included studies. Twelve out of these 13 studies provided some evidence on the consequence of exposure to gun violence on social conduct or disruptive behavior among children and adolescents (Beardslee et al., 2018; Beardslee J et al., 2021; Bingenheimer

et al., 2005; Comer BP & Connolly EJ, 2023; Forster et al., 2015; Gard et al., 2022; Kagawa et al., 2020; Leibbrand et al., 2020; Lennon et al., 2023; Magee et al., 2022; Reid et al., 2017; Sheley and Brewer, 1995; Shulman et al., 2021).

Of these, five studies assessed the relationship between exposure to gun violence and subsequent gun carrying. One of the earliest studies included in our review, a cross-sectional study by Sheley and Brewer (1995) found that being threatened with a gun was related to higher gun carrying behavior among both boys and girls (Sheley and Brewer, 1995). Some longitudinal studies further provided temporal direction on the relationship between gun violence and subsequent gun carrying. For example, in a study on male juvenile offenders adjudicated for a serious offense, those who were victims of gun violence were 43% more likely to engage in future gun carrying behavior (Beardslee et al., 2018). A similar longitudinal study assessed gun carrying behavior every other six months among serious offenders over the course of two years (Reid et al., 2017). During all four of these stages, the authors found that adolescents exposed to firearm violence were significantly more likely to carry guns over the period of the study. Using a sample of non-serious juvenile offenders, Beardslee et al. (2021) reported that those exposed to gun violence were over four times more likely to carry gun a future (OR = 4.25, $p < 0.001$) (Beardslee et al., 2021). Further, in longitudinal survey with a sample of over 5000 nationally representative sample, Comer and Connolly (2023) examined the relationship between early exposure to gun violence (before 12 years) and gun carrying behavior (Comer and Connolly, 2023). The authors found that early exposure was linked with increased likelihood of adolescents carrying a handgun ($\beta = 0.23$, 95% CI = 0.21 to 0.26). They also reported a greater tendency for such adolescents to carry a handgun more frequently over time ($\beta = 0.26$, 95% CI = 0.24 to 0.29), however, this relationship did not persist into adulthood.

There was also considerable evidence suggesting that exposure to gun violence increased the likelihood of other social conduct behaviors (Bingenheimer et al., 2005; Forster et al., 2015; Gard et al., 2022; Kagawa et al., 2020; Magee et al., 2022; Magee et al., 2022b; Shulman et al., 2021). In a longitudinal study of adolescents in Chicago between 1994 and 2002, Bingenheimer and colleagues showed that among adolescents within the same stratum, those exposed to firearm violence were more than twice as likely to commit a violent crime over the next two years compared to unexposed adolescents (adjusted OR = 2.47) (Bingenheimer et al., 2005). Another longitudinal study involving 3444 youth provides further evidence of association between proximity to gun violence and behavioral disorders. The authors reported a 9.6% rise in teenage delinquency for every fatal gun violence episode that happened within 500 m of a youth's home (Gard et al., 2022).

Magee et al. (2022) also reported a significant albeit small increase in Medicaid claims for disruptive behavior disorders among non-fatal firearm injury survivors (pre-shooting 6.8% vs. post-shooting 7.0%) (Magee et al., 2022). There was also evidence that gun violence exposure was significantly associated with increases in both reactive and proactive aggression (Shulman et al., 2021). The authors of this study were also able to demonstrate that this was not because of a reverse direction in the exposure/outcome relationship (i.e., aggression as a predictor of subsequent gun violence exposure).

3.3.7. Adults (n = 32)

Thirty-two studies reported a range of mental health indicators among adults, with the most common being trauma and stress related disorders (n = 20). There were also studies that reported on mood disorders (n = 12), substance-related and addictive disorders (n = 8) and disruptive, impulse-control, and conduct disorders (n = 7). Anxiety disorders were the least reported (n = 5). See Fig. 4.

3.3.8. Mood disorders (n = 12)

Mood disorders among adults were assessed in 12 studies (Fullerton et al., 2015; Galovski et al., 2023; Grieger et al., 2003; Hsu et al., 2021; Leibbrand et al., 2021; Lyons et al., 2019; Magee et al., 2022; Magee et al., 2022b; M. E. Smith et al., 2020; R. N. Smith et al., 2018; South et al., 2021; Timmer-Murillo et al., 2023). A study that assessed mental healthcare utilization 24 months before and after a non-fatal firearm injury, reported a substantial increase in the prevalence of depression among persons who sustained non-fatal firearm injuries (6.6% pre vs. 10.4% post, $p < 0.001$) (Magee et al., 2022). To determine the long-lasting impact of non-fatal firearm injuries, Smith et al. (2018) compared victims with retained bullets to those without retained bullets. The authors found that even after adjusting for factors such as severity and number of injuries, those with retained bullets experienced more severe depression symptoms ($\beta = 3.52$; $p = 0.017$) (R. N. Smith et al., 2018).

In a community-based study that assessed the prevalence of depression following a shooting spree, Fullerton et al. (2015) reported that 22% of the respondents in their study met the criteria for mild to severe depression (Fullerton et al., 2015). However, a smaller study conducted among healthcare workers within a tertiary facility within the same community by Grieger et al. (2003) shortly after the incidents found that only 8% of the participants in their study met the criteria for depression (South et al., 2021). Another study that assessed the impact of gun violence on mental health indicators across four cities found that persons who had been exposed to gun violence fatalities in their lifetime had a higher average depression score (mean = 6.06, SD = 6.06) compared to those who had not been exposed (mean = 4.89, SD = 5.55). This difference was statistically significant ($\beta = 1.25$, $p < 0.001$) (M. E. Smith et al., 2020). Similarly, Leibbrand and colleagues reported that mothers who had ever witnessed a shooting had a significantly higher depression scores compared to mothers without such an experience ($\beta = 0.214$; 95 CI: 0.133–0.294) (Leibbrand et al., 2021). Smith and colleagues reported that respondents exposed to gun violence fatalities had a higher likelihood of experiencing suicidal thoughts compared to those without such exposure (OR = 1.95, 95% CI = 1.31–2.91) (M. E. Smith et al., 2020).

On the other hand, in an analysis of a nationwide sample of veterans aged 18–50 years that oversampled residents of high crime communities, Galovski et al. (2023) found no significant difference in depressive symptoms among populations exposed to gun violence (Galovski et al., 2023). Also, Hsu et al. (2021) examined the relationship between firearm violence and suicidal ideation among young adults (18 and 24 years) experiencing homelessness in seven US cities where they found no relationship between either direct or indirect victimization and suicidal ideation (Hsu et al., 2021).

The study by South et al. (2021), which took an ecologic approach, presented mixed findings. The authors reported that emergency department complaints for depression did not significantly change before and after neighborhood shootings. However, they noted a significantly higher odds of complaints for depression (OR = 2.86, 95% CI = 1.3, 6.29) when multiple shooting incidents occurred within a six-month period within a 0.5-mile radius from the incident (South et al., 2021).

3.3.9. Anxiety disorders (n = 5)

Four out of the five studies that assessed anxiety did so among direct victims of gun violence (Haasz et al., 2023; Magee et al., 2022; Magee et al., 2022b; South et al., 2021; Timmer-Murillo et al., 2023). All four studies found a significant relationship between firearm injury and increased anxiety symptoms. Among a cohort of firearm injured patients, Timmer-Murillo and colleagues reported an increase in mean anxiety score increased from a baseline of 9.28 (SD = 9.23) to 11.20 (SD = 11.17) at 6 months. Similarly, Magee et al. (2022) also reported a significant increase in diagnoses for anxiety in the 12-months post shooting incident among both adult survivors (pre: 2.2% vs. post: 4.4%) and their adult family members (pre: 1.9% vs. post: 2.8%) (Magee et al., 2022). However, using emergency department electronic health record (EHR) data to assess the effect of community gun violence on anxiety

related complaints, South et al. (2021) found no significant increase in anxiety related complaints following incidents of community shooting (South et al., 2021).

3.3.10. Substance-related and addictive disorders (n = 8)

Eight studies examined the relationship between gun violence exposure and substance use (Fullerton et al., 2015; Ganpo-Nkwenkwa et al., 2023; Grieger et al., 2003; Lyons et al., 2019; Magee et al., 2022; Song et al., 2022a; South et al., 2021; Vella et al., 2020). Six out of these studies reported an increase in substance use among persons exposed to gun violence. In a study that assessed the long-term mental and behavioral indicators among persons who had experienced childhood violence-related injuries—majority of whom had sustained a between gunshot injury—Ganpo-Nkwenkwa and colleagues reported a higher proportion of substance use in this group (45.8%) compared to the general population (13.0%) (Ganpo-Nkwenkwa et al., 2023). Song et al. (2022b) using electronic health records of victims of nonfatal firearm injuries and their families found that firearm-injured patients had approximately 600 percent higher rates of drug use disorders in the month after injury than controls (Song et al., 2022a). This was reduced but still substantial the next year, as survivors experienced an 85% increase in drug use disorders compared to the control group. Similarly, another study found that the diagnosis of substance use disorders increased significantly from 10.5% pre-gun violence exposure to 15.5% post-gun violence exposure (Magee et al., 2022).

Vella and colleagues in prospective cohort study of firearm injured survivors observed that although individuals exposed to gun violence had lower alcohol and substance use scores, combined alcohol and substance use increased by 13.2% after gun violence exposure (Vella et al., 2020). Specifically, the percentage of participants reporting combined alcohol and substance use increased from 30.8%-pre-gun violence exposure to 44.0% post-gun violence exposure. On the other hand, South et al. (2021) did not find a significant association between gun violence exposure and substance use (South et al., 2021).

3.3.11. Trauma- and stressor-related disorders (n = 20)

With 20 studies reporting trauma and stress related symptoms among adults, this was the most featured mental health indicator in our review (Allwood et al., 2023; Aubel et al., 2021; Bailey A et al., 2013; Fullerton et al., 2015; Galovski et al., 2023; Ganpo-Nkwenkwa et al., 2023; Greenspan and Kellermann, 2002; Grieger et al., 2003; Herrera-Escobar et al., 2021; Joseph et al., 2019; Lyons et al., 2019; Magee et al., 2022; Mello-Silva et al., 2012; Nehra et al., 2021; Reese et al., 2012; Schulden et al., 2006; M. E. Smith et al., 2020; R. N. Smith et al., 2018; Timmer-Murillo et al., 2023; Vella et al., 2020).

Multiple studies consistently reported increased burden of trauma and stress related symptoms, particularly in the initial 12 months following the event (Fullerton et al., 2015; Greenspan and Kellermann, 2002; Mello-Silva et al., 2012; Nehra et al., 2021; Timmer-Murillo et al., 2023). In a national prospective study, Nehra et al. (2021) reported that following a firearm injury, PTSD symptoms notably increased (Beta = 3.82, 95% CI = 1.29–6.35) within the first year of the injury (Nehra et al., 2021). Similarly, Greenspan and Kellermann in a longitudinal study using records from subjects recruited at a Level I Trauma Center found that over 80% of victims had moderate or severe post-traumatic stress symptoms within 8 months of sustaining a firearm injury (Greenspan and Kellermann, 2002). A study from Brazil reported that 60% of firearm injury victims screened positive for PTSD after about 2 months of being discharged from the hospital (Mello-Silva et al., 2012).

A few studies also provided evidence that the risk of PTSD from firearm injury was greater compared to other forms of violent injuries. Herrera-Escobar et al. (2021) found significantly greater odds of PTSD (adjusted OR = 3.06) among victims of firearm injury compared those of motor vehicle accident (Herrera-Escobar et al., 2021). Violence involving a firearm was associated with a higher prevalence and risk of traumatic and stress related disorders compared to violence involving other forms of weapons (Allwood et al., 2023; Aubel et al., 2021). A study which compared different forms of victimization (physical, gun and knife) also showed that gun victimization more than doubled the likelihood of clinically significant PTSD symptoms (OR = 2.30) while that of knife victimization showed no significant effect (Allwood et al., 2023). Even among firearm injury victims, the burden of PTSD among firearm injury victims varied with mechanism of firearm injury. A 5-year analysis comparing readmission rate by type of firearm injury noted that while the general PTSD readmission rate for those injured by firearms stood at 9%, those injured by shotguns saw the highest readmission rate due to PTSD at 16%. Those injured by semiautomatic rifles and handguns followed with rates of 9% and 7%, respectively (Joseph et al., 2019).

Two retrospective studies provided evidence that PTSD symptoms recede with time. In a retrospective study involving adults treated during childhood for a violent traumatic injury, the authors found no significant relationship between gunshot injuries and PTSD (Ganpo-Nkwenkwa et al., 2023). This finding is comparable to that of Vella et al., (2020), who reported that gunshot victims were less likely to have a positive screen for probable PTSD five years later (Vella et al., 2020).

4. Discussion

We identified 58 peer-reviewed studies documenting the link between interpersonal gun violence and mental health. Three primary findings emerged. First, most of the studies that reported on children and adolescents found that both direct and indirect exposure to gun violence was associated with a significant increase in mental health problems especially social conduct disorders and aggression. Second, there was consistent evidence that exposure to interpersonal gun violence, particularly through direct victimization was linked to a higher risk for developing PTSD, especially among adults directly exposed. Third, proximity played a role in the impact of gun violence on mental health, with people closer to exposure incidents reporting more mental health problems than people living farther from gun violence-related incidents.

Social conduct and aggressive disorders were the most frequently reported mental health indicators among children and adolescents. We also found evidence that exposure to gun violence may be associated with behavioral changes that then increase the likelihood of future violent behaviors such as aggressions and gun carrying. This was consistently reported in five of the included studies and was the case whether the exposure to gun violence was direct or indirect. This finding aligns with the broader literature on violence and aggression which suggests that exposure to violence can lead to the normalization of violent behaviors and increase the likelihood of future violence perpetration (Widom and Wilson, 2015). This phenomenon, often referred to as the "cycle of violence" or "intergenerational transmission of violence," has been documented in various forms of violence, including domestic violence, community violence, and war-related violence (Tracy et al., 2016). In the context of gun violence, this cycle can manifest in several ways. For example, individuals who are exposed to gun violence may acquire guns themselves as a means of self-protection, thereby increasing the risk of future gun use and potential violence (Brunson et al., 2022). Further, the psychological trauma associated with exposure to gun violence may contribute to aggressive behaviors, which can further perpetuate the cycle of violence.

Another mental health indicator consistently linked to gun violence exposure was PTSD, particularly among adults directly affected by gun violence. Most of the studies in our review focused on direct victims with only a few assessing PTSD among persons indirectly exposure to gun violence. All included studies among direct victims of gun violence reported an increase in PTSD symptoms especially within a year of firearm victimization. This finding is in keeping with previous reviews that examined the effect of other forms of violence such as mass shootings on PTSD (Lowe and Galea, 2017). The review by Lowe and Galea on the mental health consequences of mass shootings found significant

increases in the prevalence of posttraumatic stress symptoms following these traumatic events. More recently, a scoping review by Ranney and colleagues, although restricted to youth and their parents, found that an increase in firearm injury was associated with high rates of posttraumatic stress symptoms (Ranney et al., 2019).

We show in this review that proximity to gun violence incidents increases the risk of adverse mental health indicators among persons affected. Several studies included in this review reported that individuals living near the site of gun violence, such as neighborhoods directly affected by shootings, experienced more adverse mental health consequences. This could be attributed to several factors such as directly witnessing the event, hearing gunshots, or heightened fear and anxiety within the immediate vicinity, all of which may heighten the traumatic experience of gun violence. Recognizing that the consequences of gun violence are closely intertwined with structural issues like racism and economic disinvestment that disproportionately affect minoritized communities (Buggs et al., 2022; Sharif et al., 2021) suggests that addressing these fundamental causes is crucial in any comprehensive approach to reduce the mental health impact of gun violence. Interventions that focus on place, such as revitalizing vacant lots and repairing abandoned homes, have shown effectiveness in decreasing gun violence in these areas (Garvin et al., 2013; South et al., 2023).

Our findings must be understood in light of limitations of this review. Most of the included studies were conducted in the US, raising questions about the generalizability of our findings to other regions with differing gun violence rates and mental health infrastructures. Further, our review did not include mass shootings and police shootings, both of which are significant and highly visible aspects of gun violence. These types of shootings can have wide-ranging impacts on mental health, not only for the victims and their families but also for entire communities and the public at large. However, they often involve unique factors and dynamics that may not be representative of gun violence more generally. As such, by focusing on more prevalent forms of gun violence, this review aimed to provide insights that are applicable to a broader range of situations and populations. We also excluded studies related to suicide (not suicidal ideation), which is a significant cause of gun-related deaths globally. Although gun-related suicides are a significant public health issue—especially in the US—they involve different dynamics and risk factors compared to other forms of gun violence. As such by excluding suicide, we were able to focus more specifically on interpersonal gun violence and its impacts on mental health.

4.1. Future research directions

This review highlights several limitations in the literature and points to the need for more research to further our understanding on the mental health consequences of gun violence. First, the relationship between gun violence and other mental health indicators beyond depression, PTSD, and anxiety remains largely unexplored. Future research should aim to investigate the impact of gun violence on a broader range of mental health indicators, including but not limited to, substance use disorders, personality disorders, and other forms of psychological distress. Understanding these relationships could provide a more comprehensive picture of the mental health impacts of gun violence and inform the development of more holistic mental health interventions for affected individuals. Second, the influence of cultural and societal contexts on the relationship between gun violence and mental health is poorly understood. Cultural and societal factors can shape individuals' experiences and interpretations of violence, their access to and use of mental health services, and their mental health indicators. Studies conducted in diverse contexts could provide valuable insights into these influences and inform the development of culturally appropriate interventions. Third, the role of individual characteristics, such as age, gender, race/ethnicity, socioeconomic status, and previous mental health conditions, in moderating the impact of gun violence on mental health is another important area for future research. Understanding these moderating factors could help to identify individuals who are at particularly high risk of adverse mental health indicators following gun violence and to develop targeted interventions for these individuals. Fourth, most of the studies identified in our review were based in the US, consistent with the predominance of gun-related violence in the US. However, other regions known for high rates of gun violence, such as Latin America and the Caribbeans (The Global Burden of Disease 2016 Injury Collaborators, 2018 Injury Collaborators, 2018), were underrepresented in the literature. Consequently, there is a need to expand research efforts to these regions to gain a more global understanding of the mental health impacts of gun violence. Last, the potential cyclical nature of gun violence, mental health, and gun carrying behaviors is an area that warrants further investigation. Understanding these complicated relationships could inform efforts to break the cycle of violence and its associated mental health impacts.

5. Conclusion

Our systematic review highlights the far-reaching consequences of gun violence, extending beyond immediate physical injury to include enduring mental health impacts. The review reveals a significant association between exposure to gun violence and adverse mental health indicators, including mood disorders such as depression, trauma and stress related disorders especially PTSD, and disruptive and conduct disorders particularly gun carrying among adolescents. The impact of gun violence on mental health appears to vary based on proximity to gun violence incidents with those closer to incidents being more predisposed to adverse mental health indicators. Overall, our findings underscore the need for comprehensive, targeted interventions and further research to better understand and address the far-reaching mental health consequences of gun violence.

CRediT authorship contribution statement

Mohammed Abba-Aji: Conceptualization, Formal analysis, Methodology, Visualization, Writing – original draft. **Shaffi Fazaludeen Koya:** Formal analysis, Methodology. **Salma M. Abdalla:** Conceptualization, Writing – review & editing. **Catherine K. Ettman:** Formal analysis, Visualization, Writing – review & editing. **Gregory Herschel Cohen:** Formal analysis, Writing – review & editing. **Sandro Galea:** Conceptualization, Formal analysis, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ssmmh.2024.100302.

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